Luis Enrique Namigtle Jiménez — Curriculum Vitae

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Date of Birth: March 06, 1997

Professional Objective

My professional goal is to be constantly updated in the area of electronics and its branches; nowadays it is important to be aware of the great scientific and technological advances that are taking place worldwide. Personally, I consider that I am a responsible person with my work, willing to contribute my knowledge and also to learn from those around me, strengthening my abilities, skills, and commitments in the area of my interest. In addition, I am focused on digital design verification using UVM methodology, developing testbenches, performing functional verification, and applying coverage-driven verification to ensure the correctness and reliability of FPGA and ASIC designs.

Main fields of interest

- Design and development of FPGA architectures.
- FPGA state machine design.
- Verilog and VHDL for RTL model development, design and implementation.
- Design and implementation of filters in FPGA.
- HDL simulation with GTKWAVE and ICARUS VERILOG.
- Signal processing for biometrics applications.
- Application of virtual instrumentation in LabVIEW.
- Verification of digital designs using UVM methodology.
- Debugging and analysis of simulation results using UVM reports and waveforms with Synopsys and Cadence tools.
- Development of Python scripts to automate UVM testbench templates and verification environments for Synopsys, Cadence, and Xilinx Vivado.
- Makefile manipulation for toolchain configuration and automation across Synopsys, Cadence, and Vivado.
- Application of SystemVerilog DPI for C/Python integration in verification flows.
- Digital Mixed-Signal (DMS) modeling of analog circuits applied to UVM verification.
- Collaborative verification workflows using GitHub for version control and team integration.

Labor background

- **QSM Semiconductores** — **Corregidora, Querétaro Arteaga, México**
  Digital Design Electronics Engineer. — *January 2024 – December 2025*

  In my current position, I am in charge of the design of descriptive hardware languages (VHDL) and Verilog, where I implement architectures for FPGA boards. In addition, I use software to verify the functionality of the architectures and develop test benches to ensure that the designs meet the established objectives.
  I worked with ASICs and FPGAs from Renesas' ForgeFPGA family, focused on low-density and low-power solutions. I was responsible for designing architectures to validate the functionality and performance of these devices.

  - Elaboration of state machines in VHDL.
  - Testbench design in VHDL.
  - Programming for PAL and GAL gates.
  - Linux operating system manipulation

- ROS and microsRos manipulation in embedded systems applications : Raspberry pi pico.
- Testbench creation for IP core. I have used tools such as QuestaSim to design and simulate Testbench for the verification of RTL architectures.
- Simulation with the OpenLane tool to design the integrated circuit.

- **Flextronics Manufacturing Mex, S.A. de C.V** — **Guadalajara, Jalisco**
  Failure Analysis Technician, Grade 08. — *September 2023 – December 2023*

  I was responsible for the review of telecommunication boards at CIENA. In this position, I used various tools, such as multimeter, oscilloscope and power supplies, to perform circuit analysis and detect faults accurately. In addition, I have experience in handling the company's internal software, such as Linux, View Expert, View Test, Xilinx and Quartus, the latter two allowed me to efficiently implement '.bit' files on their boards.

  - Diagnostic Technician on telecommunication cards for CIENA.
  - Failure analysis use of measuring instruments such as multimeters, oscilloscopes, power supplies, etc.
  - Management of Linux to load test files on the cards.

Academic Background

Instituto Nacional de Astrofísica, Óptica y Electrónica, INAOE
Master's degree in Electronics Sciences, specializing in Instrumentation FPGA.m.
Thesis topic: Architectures PUFs in FPGA for cancellable biometrics.
Period: August 2020 - March 2023.
Professional license: 13498080

Benemérita Universidad Autónoma de Puebla, BUAP
Bachelor's degree in Electronics, specializing in Optics and Instrumentation.
Period: August 2015 - November 2020.
Professional license: 12257195

Relevant projects

- **PhD:** *Verification of ASIC using SystemVerilog and UVM*
  - Development of Python scripts to automate UVM testbench templates and verification environments for Synopsys, Cadence, and Xilinx Vivado.
  - Manipulation of Makefiles for efficient configuration and toolchain integration across multiple EDA tools.
  - Application of SystemVerilog DPI (Direct Programming Interface) to extend verification capabilities through C and Python integration.
  - Use of Digital Mixed-Signal (DMS) modeling to represent analog circuit behavior in UVM-based verification.
  - Collaboration with research teams using GitHub for version control and teamwork in verification projects.
- **Proyecto Lagarto:** *Verification of Digital Designs using SystemVerilog and UVM*
  - Gained hands-on experience in functional verification using SystemVerilog and UVM methodology.
  - Developed testbenches in EDA Playground to validate digital designs.
  - Performed verification workflows in Vivado, applying constrained-random stimulus and assertions.
  - Extended verification to professional EDA tools such as Cadence, using UVM testbenches for functional coverage and protocol compliance checking.
  - Debugged and analyzed simulation results using UVM reports and waveform viewers, ensuring design correctness and reliability.
- **Bootcamp:** *Digital Integrated Circuits Design Bootcamp*
  - Hands-on experience using and implementing Synopsys tools for digital design.
  - Designed a simple CPU architecture using Verilog HDL.

- Developed and implemented a functional testbench to verify CPU behavior.
- Gained familiarity with **Design Compiler (DC) Shell** synthesis flow: RTL reading and linking, constraints setup, pre-compile reports, compilation/synthesis, post-compile reports, and design saving.
- Performed full synthesis of the CPU using RTL dataflow methodology.
- Applied advanced synthesis techniques including clock gating, register optimization, incremental compilation, area optimization, and datapath extraction using DesignWare components.

- **Master's Project:** *'Architecture PUF in FPGA for cancellable biometrics.'*
  - Manipulation of primitive functions for Spartan 6 board.
  - Placement and routing of components.
  - Design and implementation of PUF architectures with manual and random routing.
  - Design and implementation of RS-232 protocol.
  - Design of a state machine for FPGA-Computer communication control through RS-232 protocol.
  - Simulation of cancellable biometrics in MATLAB.
- **Master's Project:** *'Training of a Neural network for a nonlinear system using the perceptron multilayer method.'*
  - Identification of a non-linear system (Beam and Ball).
  - Training and validation of 1000 data of the nonlinear system.
  - Simulation of the neural network in MATLAB.
- **Master's Project:** *'Filtering and implementation of an ECG signal FPGA.'*
  - MATLAB FIR filter simulation to remove: DC component, 60 Hz component and Frequencies above 10 Hz.
  - Implementation of the FIR filter coefficients with a 32 bits fixed point format on the FPGA.
- **Master's Project:** *'Design of a RAT RACE.'*
  - Design of a RAT RACE with FR-4 substrate and a frequency of 2.4GHz using Advanced Design System(ADS) Software.
- **Master's Project :** *'Implementation of a Physical Unclonable Function: SRAM in FPGA.'*
  - Manipulation of primitive functions for Spartan 6 board.
  - Placement and routing of components.
  - Implementation of SRAM architecture through ISE Xilinx .
- **Master's Project:** *'Design of laboratory practices for the DE2-Cyclone II(Altera) board using the FPGA Processor architecture. '*
  - Design and implementation of practices through Quartus Software.
  - Application NIOS II.
  - Adaptation of DE0-Nano board practices to DE2 board.
- **VIEP(Vicerrectoria de investigación y Estudios de Posgrado)/BUAP Project :** *'ELASTIC LIGHT SCATTERING THROUGH POLYSTYRENE PARTICLES'*.
  - Obtaining of polystyrene powder particles.
  - Measurement of the particles with atomic force microscope(AMF).
  - Measurement of the particles with ultraviolet visible electroscopy(UV-VIS).
  - Lens arrangement for beam collimation.
- **VIEP/BUAP Project :** *'Manipulation and characterization of Polystyrene particles'*.
  - Search and review specialized information on the optical technique(s) to be used.
  - Preparation of samples depending on the optical characterization technique to be used.
  - Processing and analysis of experimental results.
- **Optoelectronics degree/BUAP Project:** *'Development of a current-controlled voltage source through LabVIEW for the control of a photomultiplier(PMT).'*
  - Graphical interface design through LabVIEW.
  - GUI Implementation in FPGA Spartan 6 board.

- Implementation on a USB-1024LS Data Acquisition card with 24 Inputs.

- **Professional internship/BUAP:** *'Implementation of a system for the use of microparticles in scattering.'*

Articles

- Namigtle-Jiménez, A., Bautista-Merino, O., Namigtle-Jiménez, J., Cortés-Vázquez, O., & Namigtle-Jiménez, L. E. (2022). MPC implementation in an heat exchanger using LabVIEW. Pädi Boletín Científico de Ciencias Básicas e Ingenierías del ICBI.
- Namigtle J., J., Alvarado V. M., Namigtle J., A., Namigtle J., L. E., Enriquez G. M., Bautista M. O., Mastache M. J. E., Accuracy and Computational Cost of the Hammerstein-Wiener Structure Applied to the Nonlinear Systems Identification. On review. IEEE Latin American Transactions (2022)

Academic productivity

Courses, conferences and/or prizes

- Certificate of attendance for the conference: Verification Methodology for the Development of Application-Specific Integrated Circuits (ASIC). INAOE 2025.
- National Semiconductor Summer School. "Proyecto Lagarto". IPN 2025.
- Python Course. Santander 2025.
- Language: SystemVerilog Assertions for formal verification. Synopsys 2025.
- VCS RTL and Gate level simulation. Synopsys 2025.
- RTL architect: Jumpstart. Synopsys 2025.
- Taller de tecnología de semiconductores, INAOE y MOS-AK. INAOE 2025.
- Digital Integrated Circuits Design Bootcamp. Synopsys 2025.
- National Semiconductor Summer School. "Proyecto Lagarto". IPN 2024.
- English for the Semiconductor Industry. TecNM 2024.
- Semiconductor Diploma. TecNM 2023.
- Meet Linux. Computational Intelligence. INAOE ,UAM, Tec Monterrey 2022.
- Seminar and National School of Learning and Computational Intelligence.INAOE 2022.
- Elastic light scattering through polystyrene particles. BUAP/VIEP 2019.

Technical and personal skills

- **Programming Languages:** C++, Matlab, Scilab, TeX, VHDL, Verilog, Python, SystemVerilog, UVM, Markdown.
- **Specialty Software:** Matlab, GNU Octave, Simulink, LabVIEW, Aldec Active, SolidWorks, Qucs, Xilinx ISE, Edaplayground, Quartus, Vivado, Maple, ADS (Advanced Design System), CIMCO, Ansys HFSS, PSpice, Step7, Office, Github, GitLab, Beyond Compare, QuestaSim, Icarus Verilog, ModelSim, Go Configure Software Hub (ForgeFPGA), Synopsys VCS, Synopsys Verdi, Cadence Xrun, Cadence Virtuoso, Cadence Formal, Jira.
- **Open-Source EDA Tools:** OpenLane 1.0/2.0, Magic VLSI, KLayout, Netgen, Yosys, Sky130 PDK, GDSViewer, GHDL, GTKWave.
- **Operating System:** Windows, MAC OS, LINUX (Ubuntu 18.04, Mint 21.3).
- **Language:** Intermediate English. Certification TOEFL(550).
- **Virtues:** Teamwork, responsibility, leader, self-starter and proactive.